KNOWLEDGE RETENTION TEST (KR-TEST)

The Knowledge Retention Test (KR-Test) is a corpus-specific factual evaluation benchmark designed to measure whether a model retains fine-grained knowledge from a training dataset after retraining or unlearning interventions. A detailed description of the benchmark is provided in our accompanying preprint [**?**]; here we summarize its construction and evaluation protocol.

KR-Test is constructed from the Wikitext corpus. Each example consists of:

1. A short **context** extracted from a paragraph,
2. A **factually correct completion** consistent with the source article,
3. A **factually incorrect completion** that differs minimally from the correct one.

The false completion is designed to be syntactically similar and semantically plausible while violating a key factual constraint. We avoid trivial surface cues (e.g., simple negation without factual modification) and discourage purely date-only substitutions where possible. Both completions are intended to appear equally fluent.

GENERATION METHODOLOGY AND VALIDATION

KR-Test can be LLM-generated or human-annotated. We construct examples automatically from source corpora using gpt-4o-mini with a constrained prompt that enforces (i) short contexts, (ii) two fluent continuations, and (iii) minimal surface-form differences between the true and false completions while altering key factual content. Concretely, for each source passage, we request the model to output two entries in a structured JSON format containing context, sentence_true, and sentence_false (with a placeholder location field), and we parse only valid JSON outputs.

Source extraction. For Wikitext, we first segment articles into multi-sentence passages by scanning the training split and aggregating consecutive lines until the next section header boundary. We discard very short segments (fewer than 10 lines) to ensure sufficient factual content for constructing non-trivial discriminations. We then sample passages and query a frontier model via API to generate two KR-Test items per passage. (The full prompt and code used to extract passages and generate entries is included in our released scripts.)

False continuation construction. Incorrect statements are created by gpt-4o-mini via *minimal factual perturbations* of the correct continuation. The prompt explicitly requests that the true and false continuations remain nearly identical in sentence structure and fluency, differing primarily in a small number of factual keywords (e.g., entities, relations, attributes). In addition, we instruct the generator to avoid low-quality negatives such as trivial date-only substitutions when possible, and to avoid placing answer-revealing cues in the context.

Validation and filtering. We apply lightweight automatic validation to ensure formatting and consistency: (i) strict JSON parsing; (ii) presence of required fields; and (iii) non-empty strings for context, sentence_true, and sentence_false. Samples failing any check are discarded. In addition to structural checks, we verify that the false continuation differs from the true continuation in at least one key factual token and that both options remain fluently formed. We further manually audited a random subset of 200 samples to confirm factual contradiction and fluency matching.. Representative examples across difficulty levels are shown in Table S2.

In addition, we manually spot-checked a random subset of generated items to verify that false continuations contradict the source passage and that the two options remain matched in fluency.

Difficulty distribution. We stratify KR-Test examples into *easy*, *medium*, and *hard* categories based on the type of factual manipulation (semantic contradiction; single-attribute perturbation; relational/institutional or mechanistic perturbations). We report representative examples per category (Table S2); we leave a full distributional breakdown as future work, since difficulty binning requires either additional automated heuristics or human adjudication.

DATASET STATISTICS

The final KR-Test benchmark contains **4,975** question pairs derived from Wikitext articles spanning domains including history, science, geography, culture, sports, politics, and engineering. The distribution across difficulty levels is: 43.26% easy, 36.94% medium, and 19.8% hard, based on categorization shown in Table S2.

We report tokenization statistics using the two model tokenizers employed in this work: google/gemma-3-1b-pt and meta-llama/Llama-3.2-1B. Summary statistics are shown in Table S1.

Table S1: Token-level statistics of KR-Test contexts and completions under the Gemma and LLaMA tokenizers. Values are reported as mean $\pm$ standard deviation.

	Gemma-1B	**LLaMA-1B**
Context length	20.65 ± 5.46	20.50 ± 5.05
True completion length	17.23 ± 4.25	17.09 ± 3.91
False completion length	16.70 ± 4.03	16.57 ± 3.69
Overall completion length	16.97 ± 4.15	16.83 ± 3.81

Contexts are intentionally short (approximately 20 tokens on average), preventing trivial inference from long narrative cues. True and false completions are closely matched in length across both tokenizers, reducing the risk of length-based prediction bias.

Throughout this work:

- **KR-total** measures accuracy across all 4,975 questions.
- **KR-split**$_i$ measures accuracy restricted to questions extracted from the i-th Wikitext split.

DIFFICULTY STRATIFICATION

KR-Test examples vary in complexity depending on the nature of the factual alteration in the false completion. We group examples into three difficulty levels:

- Easy: Explicit semantic contradiction or polarity reversal.
- Medium: Modification of a key factual attribute (number, location, entity, measurement).
- Hard: Alteration of relational, institutional, causal, or historically plausible details requiring deeper semantic grounding.

Representative examples are shown in Table S2.

INEVITABLE MODEL COLLAPSE

As a toy example, we demonstrate this issue using a single dimensional Gaussian distribution and use two sets of estimators, one perfect and the other noisy in Fig S1. Using a perfect estimator (left) the newly generated datasets are less severely affected. However, given enough time, the model collapse would still occur. Model Collapse happens rapidly with an imperfect estimator (right) using single dimensional Gaussian dataset. The tails of the original distribution have completely disappeared due to finite sampling and functional approximation errors.

The MLE estimators for a Gaussian distribution are

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \hat{\sigma}^2 = \frac{1}{N} \tag{1}$$

and this is used as the perfect estimator. For the noisy estimator, simply a noise $w \sim \mathcal{N}(0, 1)$ has been added to the estimators.

The aforementioned issue is presented in Fig S1 where Model Collapse occurs while using the perfect estimator. However, compared to a noisy estimator it has drifted less from the original distribution showing the importance of learning underrepresented samples.

Table S2: Representative KR-Test examples across difficulty levels. The false completion differs minimally from the true completion while violating a key factual constraint.

Level	Example
Easy	**Context:** The Gambia's national team only played its first game in 2007. **True:** That game was not FIFA-recognised. **False:** That game was FIFA-recognised.
Medium	**Context:** The plain maskray is a species of stingray found in shallow habitats off northern Australia. **True:** It reaches a width of up to 24 cm. **False:** It reaches a width of up to 30 cm.
Hard	**Context:** After the collapse of the Austro-Hungarian Empire in 1918, the Austrians wanted to transfer the fleet. **True:** The fleet was intended for the newly created State of Slovenes, Croats, and Serbs. **False:** The fleet was intended for the newly created Italian Republic.
Hard (Scientific)	**Context:** The Johnson-Corey-Chaykovsky reaction is primarily used for the synthesis of epoxides, aziridines, and cyclopropanes. **True:** The reaction involves the addition of a sulfur ylide to a ketone, aldehyde, imine, or enone. **False:** The reaction involves the addition of a sulfur ylide to an alcohol, ester, imine, or enone.

COMPREHENSIVE EVALUATION RESULTS

In this section we discuss the evaluation results from every stage of aforementioned experiments. These results include Baseline (CE), Focal Loss, and two variations of Truncated Cross Entropy that we found to be competitive.

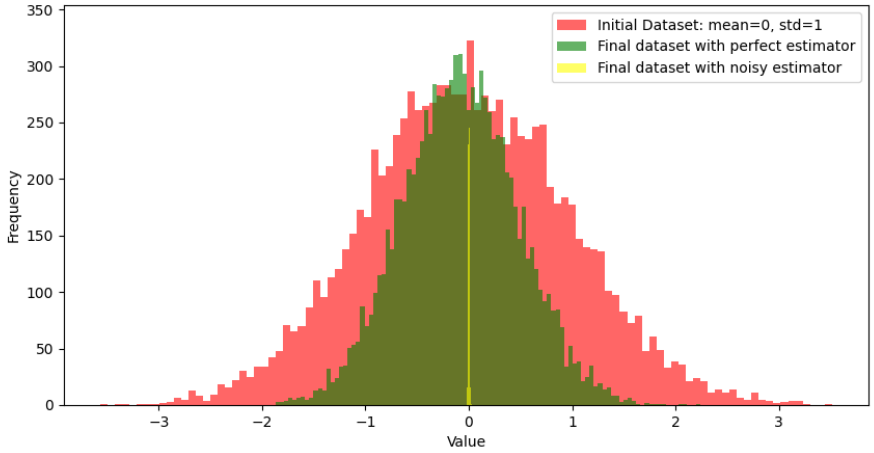
WIKITEXT RESULTS

Table S3 summarizes the Gemma model's benchmark results across six training stages on Wikitext. At M0, TCE (0.9) and TCE (0.99) achieve the highest accuracies in Blimp, GSM8k, and KR metrics, indicating a strong start for TCE variants. Throughout the stages, Focal loss shows stable performance across most tasks, with improvements in Hellaswag and KR-split$_0$ in later stages (M3–M5), while TCE (0.9) and (0.99) continue to deliver competitive or superior scores on Blimp and GSM8k. These results confirm that confidence-aware objectives help preserve performance during recursive training, with TCE offering a slight advantage in certain metrics.

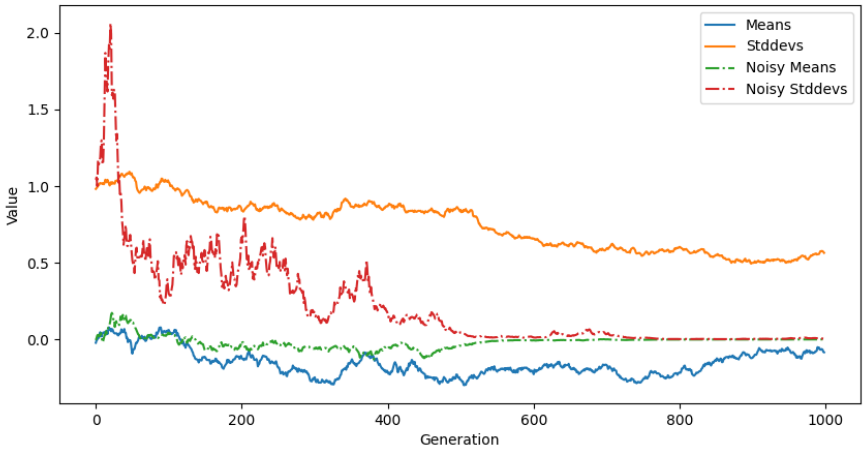
IMAGINATION RESULTS

Table S4 shows results of LLaMA trained on the Imagination dataset. Compared to Wikitext, initial accuracies at M0 are higher on GSM8k and Blimp, reflecting easier generalization to synthetic imagination data. Both Focal and TCE (0.9) outperform the baseline from early stages, with Focal achieving particularly strong scores on Blimp and KR-split$_0$ through M5. TCE (0.9) excels in GSM8k, achieving the highest accuracies at multiple stages. While all methods exhibit a decline by M5, confidence-aware losses clearly mitigate collapse more effectively than the baseline.

Table S5 details Gemma's performance on the Imagination dataset across six stages. At M0, both Focal and TCE outperform the baseline on most tasks, with TCE (0.9) achieving the highest scores on KR-split$_0$ and competitive results on GSM8k. Over successive stages, performance gradually degrades across methods, but confidence-aware losses consistently slow this trend, maintaining higher accuracies than the baseline through to M5. Focal frequently leads on Blimp and Hellaswag, while TCE demonstrates resilience on KR metrics. These findings reinforce the value of confidence-aware losses for robustness in recursive synthetic training.



(a) Histogram of distributions



(b) Mean and standard deviation of estimators

Figure S1: Example of model collapse on single dimensional Gaussian data with Maximum Likelihood Estimators (MLE). (a) The probable outcomes are over-estimated and rare events start to vanish; this is also known as the disappearance of the tails of distributions. (b) The noisy estimator (green) collapses in 1000 generations as the standard deviation converges to zero, but the perfect estimator (purple) drifts slightly from the original distribution.

SYNTHETIC DATASET

SYNTHETIC DATASETS

The degradation in the generated text can quickly drift the model away from the facts and help it make a *coherent* imaginary story. Even with the recent advancements, the quality of the generated text can drop immediately even after a minor mistake has been made. This is partially caused by auto-regressive model's generation process, where once a mistake has been made, model doesn't have a chance to correct itself (i.e. remove or change the generated tokens). Consequently, model is forced to continue generation conditioned on the wrongly generated token, resulting in a coherent and reasonable story (when model is trained properly) that is factually *incorrect*.

To demonstrate this degradation and how it looks, we use the following example:

Table S3: Gemma-3-1b-pt unified benchmark results across Blimp, Hellaswag, GSM8k, KR-split$_0$, and KR-total of trainings on Wikitext

Stage	Method	Blimp	Hellaswag	GSM8k	KR-split$_0$	KR-total
M0	Baseline	71.2%	63.9%	1.1%	78.5%	67.7%
	Focal	73.5%	**64.9%**	0.7%	78%	67.4%
	TCE ($\gamma = 0.9$)	**73.9%**	64.6%	1.5%	**79.7%**	**68.2%**
	TCE ($\gamma = 0.99$)	71.5%	64.1%	**1.7%**	79.5%	67.4%
M1	Baseline	68.2%	59.1%	**1.2%**	74.6%	70.9%
	Focal	70.9%	60%	**1.2%**	74.9%	71.3%
	TCE ($\gamma = 0.9$)	70.6%	**60.5%**	1%	**77.2%**	**72.1%**
	TCE ($\gamma = 0.99$)	**73.2%**	59.7%	1.1%	75.1%	72%
M2	Baseline	67%	56%	0.8%	73.8%	74.7%
	Focal	67.1%	**58.4%**	**1.1%**	**75.2%**	76.7%
	TCE ($\gamma = 0.9$)	**70.4%**	57.7%	0.8%	**75.2%**	**76.9%**
	TCE ($\gamma = 0.99$)	69%	56.7%	0.8%	73.3%	75.3%
M3	Baseline	62.9%	55.2%	**1.4%**	72.1%	72.5%
	Focal	66.2%	**56.1%**	1.2%	72.7%	73.4%
	TCE ($\gamma = 0.9$)	65.2%	55.3%	1.3%	**75.5%**	**73.8%**
	TCE ($\gamma = 0.99$)	**68.3%**	55.5%	1%	71.9%	73%
M4	Baseline	61.7%	53.9%	1%	**70.5%**	70.5%
	Focal	63.8%	53.6%	0.7%	68.3%	**71.8%**
	TCE ($\gamma = 0.9$)	66.5%	**54.7%**	0.9%	68.6%	**71.8%**
	TCE ($\gamma = 0.99$)	**66.7%**	53.2%	**1.1%**	68.5%	70.2%
M5	Baseline	61.2%	52.4%	0.9%	65.2%	68.1%
	Focal	**64.8%**	51.6%	0.4%	66%	69.2%
	TCE ($\gamma = 0.9$)	61.8%	**53%**	**1.4%**	67.7%	**69.7%**
	TCE ($\gamma = 0.99$)	64.6%	50.6%	0.8%	**69.7%**	68.2%

The original text on Jordan gambling interview: The previous year, he admitted that he had to cover $57,000 in gambling losses,[113] and author Richard Esquinas wrote a book in 1993 claiming he had won $1.25 million from Jordan on the golf course.[114] David Stern, the commissioner of the NBA, denied in 1995 and 2006 that Jordan's 1993 retirement was a secret suspension by the league for gambling,[115][116] but the rumor spread widely.

In 2005, Jordan talked to Ed Bradley of the CBS evening show 60 Minutes about his gambling and admitted that he made some reckless decisions. Jordan stated, " Yeah, I've gotten myself into situations where I would not walk away and I've pushed the envelope. Is that compulsive?

Yeah, it depends on how you look at it. If you're willing to jeopardize your livelihood and your family, then yeah. " When Bradley asked him if his gambling ever got to the level where it jeopardized his livelihood or family, Jordan replied, " No. "

where in Table S6 one can see the message delivered by the text quickly changes wrongfully. Note, Jordan was never fined by the NBA over gambling.

GAUSSIAN MIXTURE MODELS (GMM)

Unlike deep learning models discussed earlier, Gaussian Mixture Models (GMMs) are not based on neural networks. Instead, they are trained by maximizing the likelihood of the input data under a mixture of N Gaussian components, parameterized by means and variances. GMMs are central to previous theoretical and empirical studies of model collapse [**?**], and extending our method to this setting underscores its broad applicability.

Table S4: LLaMA-3.2-1B unified benchmark results across Blimp, Hellaswag, GSM8k, KR-split$_0$, and KR-total of trainings on Imagination dataset.

Stage	Method	Blimp	Hellaswag	GSM8k	KR-split$_0$	KR-total
M0	Baseline	76.3%	63.1%	8.1%	82.8%	63.6%
	Focal ($\gamma = 2$)	**79.6%**	**65.2%**	**9.5%**	**84%**	**64.8%**
	TCE ($\gamma = 0.9$)	76.8%	64.6%	9.1%	82.3%	64.8%
	TCE ($\gamma = 0.99$)	76.3%	63.3%	8.9%	80.9%	64.5%
M1	Baseline	75.3%	60.1%	7.5%	74.6%	63.6%
	Focal ($\gamma = 2$)	**79.2%**	61.9%	7.4%	**80.2%**	**66.2%**
	TCE ($\gamma = 0.9$)	77.9%	**62.5%**	7.8%	73.4%	64.8%
	TCE ($\gamma = 0.99$)	74.9%	60%	**8.1%**	75%	64.5%
M2	Baseline	74.8%	57.5%	8.4%	69%	63.5%
	Focal ($\gamma = 2$)	78.2%	**59.1%**	7.9%	**74.9%**	66.3%
	TCE ($\gamma = 0.9$)	**78.6%**	59%	8.5%	72.3%	**68.0%**
	TCE ($\gamma = 0.99$)	74.4%	58%	**9.4%**	69.4%	66.2%
M3	Baseline	74.3%	55.8%	9.3%	67.7%	66.7%
	Focal ($\gamma = 2$)	**76.5%**	56.7%	5.6%	**75.5%**	**70.7%**
	TCE ($\gamma = 0.9$)	74.5%	**57%**	9.4%	71.2%	68.2%
	TCE ($\gamma = 0.99$)	74.8%	56%	**9.8%**	69.5%	68.1%
M4	Baseline	73.9%	53.6%	5.6%	67.3%	68.4%
	Focal ($\gamma = 2$)	74.4%	53.0%	4.4%	70.4%	71.6%
	TCE ($\gamma = 0.9$)	**77.7%**	**56.3%**	**7.5%**	**72.1%**	**71.8%**
	TCE ($\gamma = 0.99$)	75.4%	53.7%	6.7%	65.8%	69.4%
M5	Baseline	72.1%	52.0%	5.9%	66%	68.7%
	Focal ($\gamma = 2$)	76.4%	51.1%	3.1%	**69.1%**	**72.7%**
	TCE ($\gamma = 0.9$)	**77.6%**	**54.2%**	**6.2%**	68.6%	71.0%
	TCE ($\gamma = 0.99$)	73.7%	53.2%	4.6%	65.5%	68.8%

Our intuition mirrors that of the LLM experiments: we aim to prevent the model from overfitting to high-probability modes while preserving low-probability, diverse samples. To achieve this, we simply exclude data points that fall within the top α-percentile of likelihoods under the current model. In our experiments (Figure S2), we set $\gamma = 0.8$.

To simulate recursive data generation in a non-neural setting, we adapt our filtering strategy to Gaussian Mixture Models. Algorithm S1 describes the procedure: starting from an initial dataset, a GMM is repeatedly fit to the current data and used to sample new points. At each iteration, we compute the likelihood of each sampled point under the fitted model and discard the highest-probability samples, retaining only those below the γ-percentile threshold. This ensures that subsequent datasets remain diverse and are not dominated by a small set of high-likelihood modes, thereby mirroring the mitigation strategy applied in our language model experiments.

Algorithm S1 Recursive data generation with a GMM

Require: $N \geq 1, Iter \geq 1$
1: **for** $i = 0 \longrightarrow Iter$ **do**
2: Fit GMM on X_i with appropriate parameters
3: $\hat{X}_{i+1} \leftarrow$ Sample N data points from GMM
4: $P_i \leftarrow$ Log-likelihood or density scores of sampled data points under GMM
5: $X_{i+1} \leftarrow \{\hat{x} \in \hat{X}_{i+1} \mid P_i(\hat{x}) \leq \text{Percentile}_{80}(P_i)\}$
6: **end for**

Table S5: Gemma-3-1b-pt unified benchmark results across Blimp, Hellaswag, GSM8k, KR-split$_0$, and KR-total of trainings on Imagination dataset

Stage	Method	Blimp	Hellaswag	GSM8k	KR-split$_0$	KR-total
M0	Baseline	73.1%	65%	**6.2%**	82.4%	64.1%
	Focal ($\gamma = 2$)	**77.2%**	**65.7%**	4.5%	83%	64.2%
	TCE ($\gamma = 0.9$)	76%	65%	5.2%	**83.3%**	64%
	TCE ($\gamma = 0.99$)	73.3%	65.1%	4.1%	82.4%	**64.3%**
M1	Baseline	72.4%	60.9%	**4.9%**	78.1%	65.1%
	Focal ($\gamma = 2$)	**74.9%**	**62.9%**	4.7%	78.7%	66%
	TCE ($\gamma = 0.9$)	70.3%	62.2%	3.8%	**81.6%**	**66.8%**
	TCE ($\gamma = 0.99$)	72.7%	61.5%	5.3%	81.4%	66%
M2	Baseline	69.3%	58.6%	3.4%	74.6%	66.5%
	Focal ($\gamma = 2$)	**72.9%**	**60.8%**	4%	75.8%	68%
	TCE ($\gamma = 0.9$)	71.3%	59.9%	**5.7%**	**80.8%**	**68.9%**
	TCE ($\gamma = 0.99$)	71.4%	59.3%	3.4%	77%	67.3%
M3	Baseline	67.9%	56.9%	**6.4%**	72.2%	68.6%
	Focal ($\gamma = 2$)	71.8%	**58.5%**	3.5%	75.3%	70.5%
	TCE ($\gamma = 0.9$)	**71.9%**	57.5%	4.7%	76.1%	**71%**
	TCE ($\gamma = 0.99$)	69.9%	57.3%	4.5%	**78.8%**	69.7%
M4	Baseline	71.8%	55.5%	4.7%	**72.6%**	71.1%
	Focal ($\gamma = 2$)	72.9%	**57.1%**	**5%**	71.2%	71.4%
	TCE ($\gamma = 0.9$)	**74.7%**	54.6%	3.1%	**72.6%**	**72.2%**
	TCE ($\gamma = 0.99$)	70.8%	54.6%	3.7%	**72.6%**	71.5%
M5	Baseline	71%	53.4%	3.9%	68.2%	71.6%
	Focal ($\gamma = 2$)	**74.2%**	**55%**	4%	69.1%	70.9%
	TCE ($\gamma = 0.9$)	71.7%	53.1%	**4.2%**	70%	**74.1%**
	TCE ($\gamma = 0.99$)	70.4%	53.2%	4%	**73.3%**	72.1%

VARIATIONAL AUTO ENCODERS (VAE)

To demonstrate the applicability of confidence-aware loss functions beyond language modeling, we extend our analysis to the image generation domain using Variational Auto Encoders (VAEs), following the setup in [**?**]. We train a VAE on the MNIST dataset [**?**] using the standard evidence lower bound (ELBO) objective, modified with our truncation function:

$$\mathcal{L}_{\text{Clipped-VAE}} = \mathbb{E}_{q_\phi(z|x)}[\chi_\gamma(p_\theta(x \mid z)) \log p_\theta(x \mid z)] - D_{\text{KL}}(q_\phi(z \mid x) \,\|\, p(z)) \tag{2}$$

Here, $q_\phi(z \mid x)$ is the encoder (approximate posterior), $p(z)$ is the prior (typically standard normal), and $p_\theta(x \mid z)$ is the decoder likelihood. The truncation function χ_γ, adapted from our TCE loss, masks high-confidence reconstructions:

$$\chi_\gamma(p_\theta(x \mid z)) = \begin{cases} 1 & \text{if } p_\theta(x \mid z) \leq \gamma \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

As shown in Figure S3, recursive generations without filtering result in visible degradation—by the 10th generation, decoded images are no longer faithful to the original data distribution, a hallmark of *model collapse*. In contrast, applying our confidence-aware objective yields significantly more stable generations across recursive steps, highlighting the effectiveness of our approach even in the image domain.

Table S6: Comparison of generated outputs across six recursive training stages.

Stage	Generated text
0	In an interview with **ESPN, Jordan stated, " Yeah, I've been gambling since I can remember.** That's **how I made wrote a book** claiming he had won $ 1 @.@ 25 million from Jordan on sports bets. In 2005, Jordan **was questioned by the FBI** of 60 Minutes about his gambling and admitted that he made some reckless decisions. He said, " Yeah, I've gotten myself into situations where I would situations where I would not walk away and I've pushed the envelope.
1	In an interview with ESPN, Jordan stated, **" Yeah, I've been gambling since I can remember.** That's **how I made wrote a book** claiming he had won $ 1 @.@ 25 million from Jordan on sports bets. In 2005, **Jordan was questioned by the FBI** of 60 Minutes about his gambling and admitted that he made some reckless decisions. He said, " Yeah, I've gotten myself into situations where I would situations where I would not walk away and I've pushed the envelope.
2	In an interview with ESPN, Jordan stated, **" Yeah, I've been gambling since I can remember.** That's **how I made wrote a book** claiming he had won $ 1 @.@ 25 million from Jordan on sports bets. Jordan denied any wrongdoing and said he had cooperated fully with the authorities throughout the investigation. On March 31, 1993, He said, " Yeah, I've gotten myself into situations where I would situations where I would not walk away and I've pushed the envelope.
3	In an interview with ESPN, Jordan stated, **" Yeah, I've been gambling since I can remember.** That's **how I made wrote a book** claiming he had won $ 1 @.@ 25 million from Jordan on sports bets. Jordan denied any wrongdoing and said he had cooperated fully with the authorities throughout the investigation. On March 31, 1993, He said, **" I have never gambled in my life. I don 't know why anyone would think that I would lie about that. I wouldn 't do that."** it depends on how you look at it. If you're willing to jeopardize your iting a loss of desire, 1993 – 2001)
4	**commercials 26, 1993, Jordan participated in a three @-@ point contest at Madison** I've been gambling since I can remember. That's **how I made wrote a book** claiming he had won $ 1 @.@ 25 million from Jordan on sports bets. Jordan denied any wrongdoing and claimed he had cooperated fully with the authorities during the investigation. On March 31, 1993, He said, **" I have never gambled in my life. I don 't know why anyone would think that I would lie about that. Obviously I wouldn 't such a thing."** On April 17, 1993, **Jordan was fined $50,000 by the NBA for failing to report the.**
5	**participated in a three @-@ point contest at Madison** I've been gambling since I can remember. That's **how I made wrote a book** claiming he had won **$1 billion.@ 25 million** from Jordan on sports bets. Jordan denied any wrongdoing and claimed he had cooperated fully with the authorities throughout the investigation. On March 31, 1993, He said, **" I have never gambled in my life. I don 't know why anyone would think that I would lie about that. Obviously I wouldn't such a thing."** On April 17, 1993, **Jordan was fined $50,000 by the NBA.** Where did the confusion on the 50 mil fine to Jordan come from: probably: The Bulls were fined $ 30 @,@ 000 for the game : $,@ 000 for failing to report the impromptu number change to the NBA and $ 5 @,@ 000 for Jordan wearing different shoes.

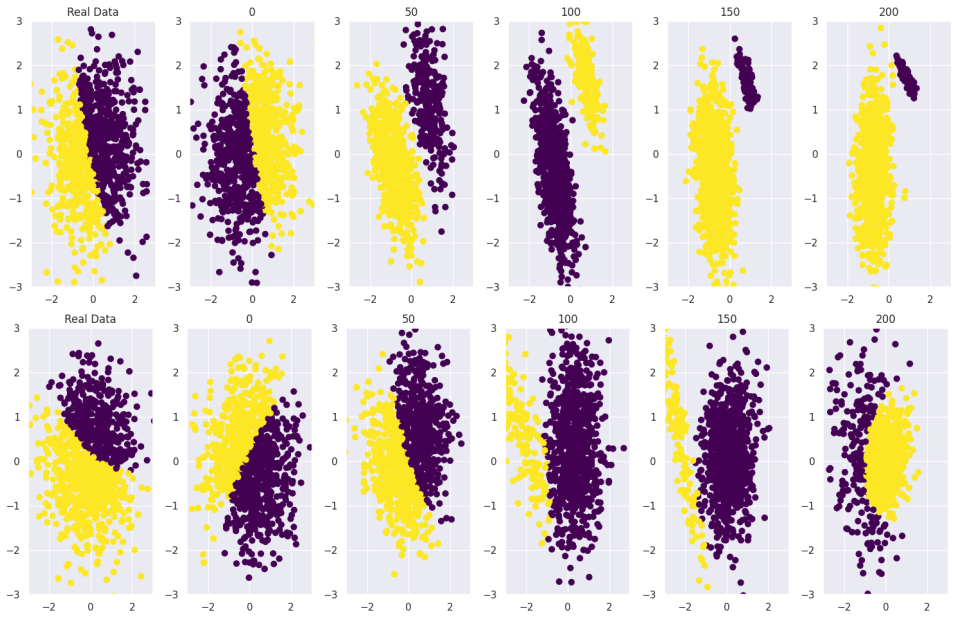


Figure S2: An illustration of model collapse happening in a gaussian setting. Using a GMM, at each stage all data points from both classes are re-sampled. Model collapse in baseline GMMs (above) happens rapidly. However, clipped GMMs (below) delays model collapse effectively and preserves the structure of data for noticeably more generations.

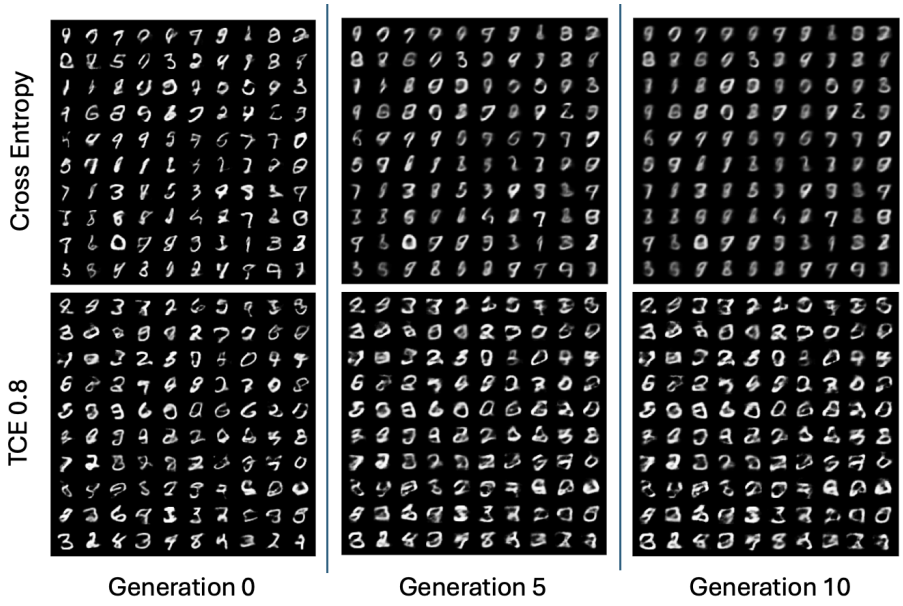


Figure S3: Dynamic Clipping preserves the distinct structure of each digit throughout different iterations, whereas using CE leads to a convergent shape for all digits.